Intercollegiate Faculty of Biotechnology
University of Gdańsk & Medical University of Gdańsk

Field of study **BIOTECHNOLOGY**

Diploma Thesis

Joanna Stacevič

Student's record number: 266026

Signature:……

Analysis of the effects of cyclophosphamide and Venetin-1 on the proteome of mouse spleen cells

Analiza wpływu cyklofosfamidu i preparatu Venetin-1 na proteom komórek śledziony myszy

This Diploma Thesis was conducted in the Core Facility Laboratories.

Signature of Supervisor: ……..

The experimental part of the thesis was performed in the Mass Spectrometry Laboratory under the supervision of dr hab. Paulina Czaplewska, prof. UG

Gdańsk 2024

Załącznik nr 1 do zarządzenia Rektora UG nr 70/R/15 ze zm.

OŚWIADCZENIE

Oświadczam, że przedłożona praca dyplomowa została przygotowana przeze mnie samodzielnie, nie narusza praw autorskich, interesów prawnych i materialnych innych osób oraz wykorzystanie materiałów wytworzonych przez generatywne narzędzia sztucznej inteligencji odbyło się w zakresie uzgodnionym z promotorem.

....

podpis *data*

Załącznik nr 3 do zarządzenia Rektora UG nr 70/R/15

OŚWIADCZENIE

Wyrażam zgodę / nie wyrażam zgody* na udostępnienie osobom zainteresowanym mojej pracy dyplomowej dla celów naukowo-badawczych.
Zgoda na udostępnienie pracy dyplomowej nie oznacza wyrażenia zgody na kopiowanie pracy dyplomowej w całości lub w części.
** niepotrzebne skreślić*

....

podpis *data*

TABLE OF CONTENTS

1. KEYWORDS
2. ABSTRACT
3. LIST OF ABBREVIATIONS
4. INTRODUCTION
   4.1 Cyclophosphamide
   4.2 Side-effects
      4.2.1 Protective agents against cyclophosphamide's toxicity
      4.2.2 Immune-stimulating agents
   4.3 Earthworms
      4.3.1 Venetin-1
5. PURPOSE OF THE PROJECT
6. MATERIALS AND METHODS
   6.1 Mice and reagents
   6.2 Protocols for CP and Venetin-1 treatment
   6.3 Homogenization of the spleen and cell lysis
   6.4 Protein digestion using FASP procedure
   6.5 Peptide purification
   6.6 Mass spectrometry analysis LC/MS-MS – SWATH-MS
   6.7 Statistical and functional analysis
7. RESULTS
   7.1 Effects of animal treatment
   7.2 Statistical analysis of SWATH-MS data
   7.3 KEGG pathways enrichment analysis
   7.4 Common altered proteins in Venetin-1 treatment groups
8. DISCUSSION
9. REFERENCES

1. KEYWORDS

SWATH-MS, proteomics, coelomic fluid, cyclophosphamide, spleen

2. ABSTRACT

Venetin-1, a protein-sugar fraction isolated from the coelomic fluid of *Dendrobaena veneta* earthworms, has previously demonstrated antitumour and antifungal properties without toxicity to normal vertebrate cells. Recent *in vitro* studies on macrophages have shown that Venetin-1 enhances the secretion of crucial multifunctional cytokines, indicating its potential for non-specific immunotherapy.

To investigate its safety and immunomodulatory effects further, *in vivo* studies were conducted on mice to confirm Venetin-1's ability to restore immune function following cyclophosphamide (CP)-induced immunosuppression. Four groups of healthy mice were selected, with two groups receiving Venetin-1 at different concentrations. Proteomic analysis was conducted on isolated spleen tissue, which plays a fundamental role in immune defence.

Global proteome changes in spleen tissue were assessed using quantitative SWATH-MS analysis. The statistical protein analysis showed minimal differences between the control group and those treated with a lower dose of Venetin-1 after immunosuppression. Meanwhile, the group treated with a higher dose of Venetin-1 exhibited significant changes in 201 proteins. Functional protein analysis revealed that many up-regulated proteins in the Venetin-1 treatment group were linked to immune and proliferation processes, such as antigen processing and presentation, platelet activation, IL-17 signalling pathway, estrogen and PI3K-Akt signalling pathways, etc. Some of these up-regulated proteins were also described in earlier publications as taking part in immune response and inflammation.

These results support the hypothesis that Venetin-1 possesses immunomodulatory properties by inducing an up-regulation of proteins involved in immune response activation. The findings suggest Venetin-1's potential application in non-specific immunotherapy, making it a promising candidate for medical preparations to treat various immunocompromised states.

3. LIST OF ABBREVIATIONS

AA - acetic acid
ACN - acetonitrile
AG - aminoguanidine
ALDH - aldehyde dehydrogenase
CAT - catalase
CCF-1 - coelomic cytolytic factor
CEN - centrifugation
CF - coelomic fluid
CKAP4 - cytoskeleton-associated protein 4
CLTC - clathrin heavy chain
COS - chitooligosaccharides
CP - cyclophosphamide
DB - 50 mM Tris-HCl pH 8,5
DDA - data-dependent acquisition
DDX21 - nucleolar RNA helicase 2
E-Syt1 - extended synaptotagmin-1
FA - formic acid
FASP - filter-aided sample preparation
FC - fold change
FDR – false discovery rate
FF - fenofibrate
FOR - formononetin
GPx - glutathione peroxidase
GSH - glutathione
GST - glutathione-S-transferase
HSC - haematopoietic stem cells
HSP90 - heat shock protein HSP 90
IAA - 55 mM iodoacetamide in UA
IFN – interferon
IFN-I - type-I interferon
IL - interleukin
Incub - incubation
LC - liquid chromatography
MANF - mesencephalic astrocyte-derived neurotrophic factor
MPO - myeloperoxidase
Myh9 - myosin-9
NaCl - sodium chloride
NER - nerolidol
NK - natural killer
PCBP-1 - poly(rC)-binding protein 1
PLEK - pleckstrin
PSG-1 - *Ganoderma atrium* polysaccharide

PSP - polysaccharides from *Polygonatum sibiricum*
ROS – reactive oxygen species
SDS - sodium dodecyl sulfate
SERPINA3K - Serine protease inhibitor A3K
SFI - Shenqi Fuzheng Injection
SFPS - water-soluble polysaccharide *Sargassum fusiform*
SWATH-MS - sequential window acquisition of all theoretical mass spectra
TNF – tumour necrosis factor
UA - 8M urea in 100 mM Tris-HCl pH 8,5
UMCS - University of Maria Składowska-Curie

4. INTRODUCTION

4.1 Cyclophosphamide

Until the 1960s, surgery and radiotherapy were the primary methods in cancer therapy. Because of the low cure rates, scientists have been trying to find better solutions. In 1943, surgeon G. Lindskog administered nitrogen mustard to his lymphoma patients, previously tested in mice by his colleagues. Three years later, Lindskog published the results of the tumour regression that he had observed. It led to the synthesis of multiple cytotoxic agents, including oral derivatives such as chlorambucil and cyclophosphamide [1].

Cyclophosphamide (CP) made its debut in 1958 as a highly valuable tool in the realms of anticancer and immunosuppressive treatments. Among diseases used to be treated by CP are chronic and acute leukaemia, multiple myeloma, lymphoma, and rheumatoid arthritis. It belongs to a group of oxazaphosphorine-substituted nitrogen mustards that function as alkylating agents [2]. The nitrogen mustards were the first compounds recognized for their clinical effectiveness against human tumours. However, they were also highly toxic to normal cells. Therefore, CP was designed to maintain a better selectivity for neoplastic cells. CP is an inactive prodrug, which was synthesized assuming that tumour cells have higher levels of phosphoramidase, an enzyme that would activate the drug by cleaving the phosphorus-nitrogen bond [3]. However, it turned out that CP is mainly activated in the liver. It requires oxidation by the cytochrome-P450 enzyme, which metabolises cyclophosphamide to its active form, phosphoramide mustard (Fig. 1) [4]. This metabolite alkylates DNA, which causes the formation of DNA-DNA cross-links, consequently resulting in DNA synthesis inhibition and cell death [5]. Another important metabolite that is produced during CP oxidation is acrolein. It inhibits the antioxidant system, producing reactive oxygen species (ROS) [6].

4.2 Side-effects

Like other alkylating agents, cyclophosphamide exhibits heightened toxicity towards rapidly dividing tissues, including the hematopoietic system, gastrointestinal tract's epithelial cells, hair follicles, and gonads [7]. The temporary side effects of CP include gastrointestinal effects (like vomiting and diarrhoea), myelosuppression, bladder cystitis, alopecia, and infection. At the same time, long-term effects include cardiotoxicity, teratogenicity, pulmonary toxicity, nephrotoxicity, and infertility [8]. Many of these toxicities are associated with acrolein, which induces the generation of reactive oxygen species (ROS). ROS cause multiple changes in the cells, including inhibiting various enzymes, induction of membrane and DNA damage, and initiation of lipid peroxidation [9] [10]. In response to oxidative stress, the JNK/c-Jun pathway is activated. As a result, c-Jun induces transcription of genes that mediate inflammatory responses like tumour necrosis factor-alpha (TNF-a). Moreover, acrolein also leads to a reduction in the activities of antioxidant enzymes such as superoxide dismutase (SOD), catalase (CAT), glutathione peroxidase (GPx), and glutathione reductase [11]. All these complications limited the practical use of cyclophosphamide in clinical settings.

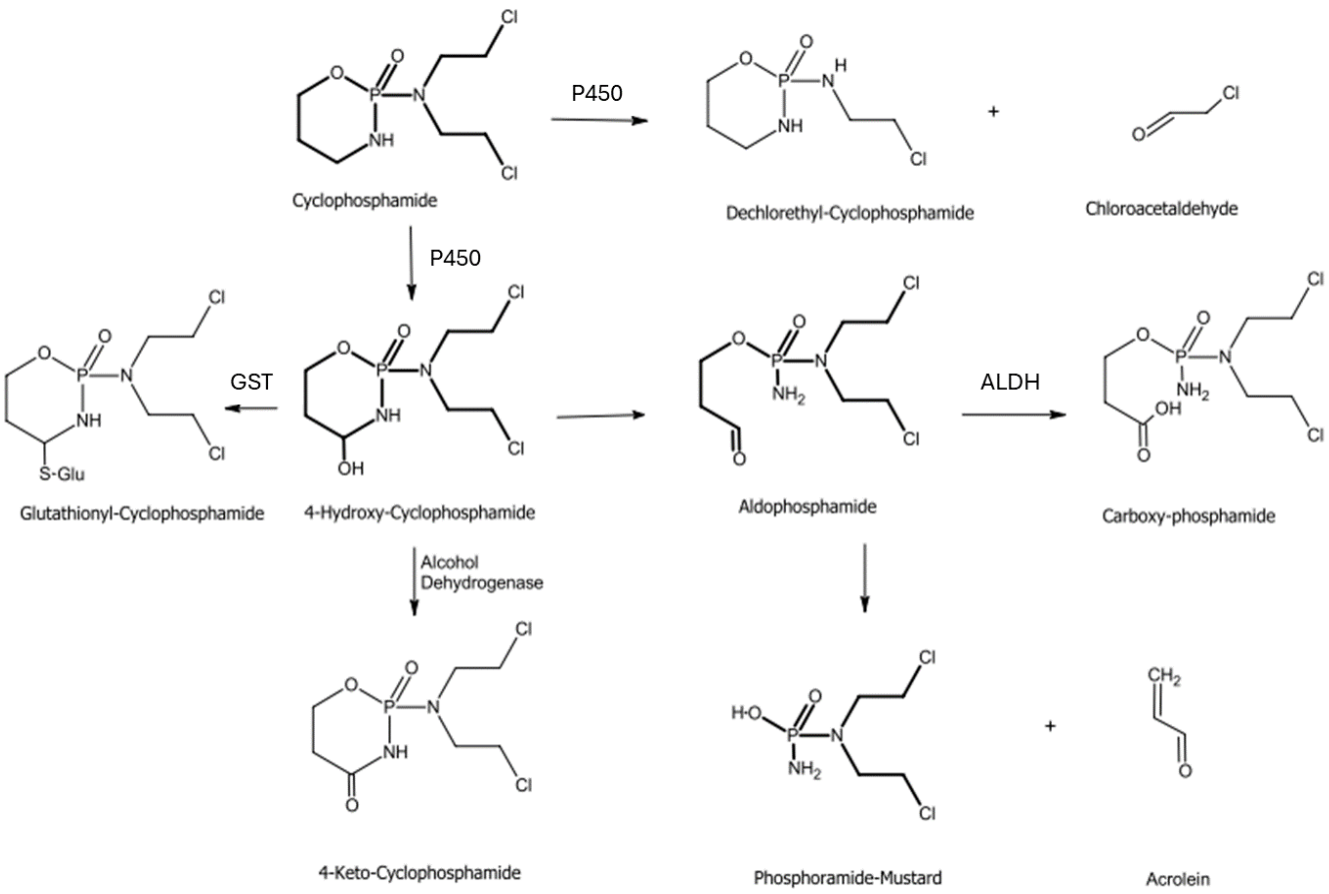

Figure 1. Metabolism of cyclophosphamide leading to the active metabolites. P450 - cytochrome P450, GST - glutathione-S-transferase, ALDH - aldehyde dehydrogenase [4].

4.2.1 Protective agents against cyclophosphamide's toxicity

Antioxidants such as flavonoids, catechins, isoflavones, anthocyanin, flavones, and isocatechins are commonly used as supplements to uphold optimal body function. Many studies have confirmed that some antioxidants have protective effects against cardiotoxicity [8], nephrotoxicity [11], hepatotoxicity [12], lung toxicity [13] [14] and some other complications caused by CP. Table 1 displays some of these attempts.

Table 1. Protective role of antioxidants against cyclophosphamide-induced cardiotoxicity [8] and nephrotoxicity [11]. GSH – glutathione; GPx - glutathione peroxidase; GST - glutathione-S-transferase; MPO - myeloperoxidase.

Animal used	Materials and methods	Results	Reference
Male Wistar albino rats	Rats were administered the same doses of corn oil (control) and probucol (61 mg/kg/day, i.p), respectively, for one week before and one week after a single dose of CP (200 mg/kg, i.p.).	Probucol prevented the development of CP-induced cardiotoxicity by a mechanism related, at least in part, to its ability to increase the mRNA expression of antioxidant genes and decrease apoptosis in cardiac tissues, consequently improving mitochondrial oxidative phosphorylation and energy production.	[15]
Female BalbC/NIH mice	Animals were treated with vitamin E (100 IU/kg, orally) 24 hr before single bolus doses of doxorubicin (10 mg/kg, intravenously) or doxorubicin and CP (150 mg/kg, i.p.).	Vitamin E in a single oral dose failed to inhibit the acute cardiotoxic activity of doxorubicin but suspended further progression of the heart muscle damage over time. On the contrary, vitamin E did not attain cardioprotection against doxorubicin and CP in combination.	[16]
Male Swiss albino mice	Animals were treated with nerolidol (NER) (200 and 400 mg/kg p.o.) and fenofibrate (FF) 80 mg/kg p.o. for 14 days, along with a single dose of CP 200 mg/kg i.p. on the 7th day.	NER 400 significantly reversed the cardiotoxic effects of CP and showed cardioprotective activity comparable with FF 80.	[17]
Male albino Wistar rats	Rats were pretreated with formononetin (FOR) (40 mg/kg/day) for 15 days, followed by CP injection on the 16th day.	Enhanced the level of antioxidants and suppressed oxidative stress. Proinflammatory mediators and apoptosis were also suppressed due to FOR treatment. The protective effect might be due to the attenuation of oxidative damage and inflammation.	[18]
Male Wistar rats	Rats were pretreated with aminoguanidine (AG) at 200 mg/kg i.p. 1 hour before administering CP at 150 mg/kg.	AG prevented lipid peroxidation, protein oxidation, depletion of reduced GSH, and loss of activities of the antioxidant enzymes, including GPx, catalase, GSTase and MPO activity. This effect might be through inhibiting oxidative stress.	[19]
Male swiss albino mice	Animals were treated with rutin (40 mg/kg and 80 mg/kg) after the administration of CP (25 mg/ kg), respectively, for 14 days.	Rutin treatment significantly reversed the status of serum biomarkers, haematological variables, and antioxidant markers.	[20]

Several natural and plant-based antioxidants have shown promising protective effects against toxicities caused by oxidative stress. However, exploring alternative solutions becomes necessary to address other side effects like immunosuppression.

4.2.2 Immune-stimulating agents

Myelosuppression is one of the most frequent struggles experienced by individuals undergoing chemotherapy for different diseases. Damage to the host defence system leads to various opportunistic diseases and infections. To overcome this problem, scientists examined numerous immune-stimulating compounds to restore immunity following CP treatment (Table 2).

Table 2. Alleviation of cyclophosphamide-induced immunosuppression on mice model organisms. IL – interleukin; IFN – interferon; TNF – tumour necrosis factor.

Animal used	Materials and methods	Results	Reference
Specific pathogen-free (SPF) Balb/c mice	Mice were treated with Shenqi Fuzheng Injection (SFI) once daily at a chosen dose (2.5 g, 5 g, or 10 g raw materials/kg) for 10 consecutive days, respectively, accompanied by intraperitoneal injection of CP (80 mg/kg) on days 4–6.	SFI treatment accelerated the dose-dependent recovery of spleen index, peripheral white blood cell, and bone marrow cell counts, enhanced T cell and B cell proliferation responses, splenic nature killer cell activity, and peritoneal macrophage phagocytosis, and restored the level of IL-2 in the serum.	[21]
Male SPF Kunming mice	The animals received chitooligosaccharides (COS) at 250, 500, and 1000 mg/kg body weight, respectively. Afterwards, they were treated with CP (60 mg/kg body weight) on days 21–30.	Thymus and spleen indices, delayed-type hypersensitivity (DTH) reaction, macrophage phagocytosis, and specific enzyme activities were significantly higher in mice treated with COS + CP than in mice treated with CP alone. Furthermore, COS treatment enhanced the production of the cytokines IL-2 and IL-12.	[22]
ICR mice	Mice from day 1 to 3 were subjected to immunosuppression by administration of CP (80 mg/kg/d) intraperitoneally. From day 4 to 10, they were administered to water-soluble polysaccharide (SFPS) isolated from Sargassum fusiforme at concentrations 50, 100 or 200 mg/kg body weight.	SFPS stimulated proliferation and the cytokines (IL2, IL-6 and IFN-) secretion of splenic lymphocytes in CP-induced immunosuppressed mice. SFPS markedly increased the phagocytic rates and cytokines (IL-2, IL-6 and TNF-) secretion of peritoneal macrophages. Moreover, SFPS significantly raised the spleen index.	[23]
Balb/c mice	Three groups of mice were treated with polysaccharides from *Polygonatum sibiricum* (PSP) with concentrations of 100, 200 or 400 mg/kg, respectively, for 10 continuous days. On the seventh day, CP (40 mg/kg, i.p.) was given to all tested groups.	Compared with the CP group, PSP accelerated recovery of spleen and thymus indexes, stimulated T cell and B cell proliferation responses, and resulted in peritoneal macrophage phagocytosis. Moreover, PSP treatment restored the levels of IL-2, TNF-α, IL-8 and IL-10 in the serum of the CP-treated mice in a dose-dependent manner.	[24]
Female BALB/c mice	Mice were treated with CP at 80 mg/kg body during the first three days. From days 4 to 10, mice were given 25, 50, or 100 mg/kg of *Ganoderma atrum* polysaccharide (PSG-1).	PSG-1 treatment fastened the recovery of peripheral red blood cells, white blood cells, and platelets and stimulated splenic natural killer cell activity and cytotoxic T lymphocyte activity.	[25]

Although multiple therapeutic approaches successfully passed *in vitro* and *in vivo* studies and even were employed in clinical practice, many of them have proven ineffective or uneconomic afterward [26].

4.3 Earthworms

Earthworms are the source of proteins, peptides, enzymes and other biologically active molecules. They play an essential role as ecosystem engineers, capable of influencing soil functions and microbial activities [27]. Besides shaping soil composition, they have been used as a traditional Chinese medicine for thousands of years. The first documented case of earthworms being used for medicinal purposes dates to 1340 [28]. Earthworms were helpful in

various ailments, such as burns, wounds, itching, and inflammation [29]. Soon, other research papers appeared, showing earthworms' antibacterial, antitumour, antifungal, hemolytic, and proteolytic activities [30]. All these biological activities are part of their cellular and humoral immunity. Earthworms synthesize various protective cells and active molecules, such as coelomocytes (leukocytes), agglutinins, lysenin, coelomic cytolytic factor (CCF-1), fetidin, etc. [29]. Coelomocytes act mainly through phagocytosis and by secreting immune-protective molecules. Agglutinins are responsible for immobilizing and aggregating foreign material [30]. Lysenin induces lysis of erythrocytes [31]. CCF-1 and fetidin were noticed to exhibit cytolytic and hemolytic properties. The bioactive substances are in a coelomic cavity filled with coelomic fluid (CF). Thus, for many years, CF has been an intriguing substance to investigate [33]. Its antibacterial, antitumour, and antifungal characteristics had great potential in improving the treatment of some diseases. However, unmodified CF is also cytotoxic against vertebrate cells.

4.3.1 Venetin-1

Venetin-1 is a protein-sugar fraction isolated from CF of earthworms *Dendrobaena veneta*. This protein-sugar fraction retains some biological activities of CF. There are already few publications presenting its antitumour effect on lung cancer cell line A549 [34], [35] and its antifungal activity on Candida albicans yeast [36], [37]. Moreover, Venetin-1 does not show cytotoxicity in normal vertebrate cells [34]. The crucial steps to modifying CF are heating it at 70 ˚C for 10 minutes and dialing it in water through a dialysis bag for 24 hours at 4˚C temperature. This way, a high-molecular-weight fraction called Venetin-1 is obtained.

There is already documented proof of its anticancer and antifungal activities, which need further investigation. On the other hand, there are also other but less studied Venetin-1 properties, which may be even more promising. In vitro studies on macrophages, performed by a group of scientists from the University of Maria Składowska-Curie (UMCS) in Lublin, showed immunostimulatory activity of modified CF fraction [38]. During this study, the mentioned fraction increased the secretion of crucial and multi-directionally acting cytokines, IL-6 and IL-1β. These findings suggest that Venetin-1 could be applicable in non-specific immunotherapy. For instance, Venetin-1 may act as an immunostimulant in adjuvants, drugs or dietary supplements used to prevent and treat many diseases.

5. PURPOSE OF THE PROJECT

The purpose of this project was to estimate the immunomodulatory properties of Venetin-1 and its effect on mice organs. Earlier, *in vitro* studies had shown that protein-sugar fraction with a molecular mass of more than 14 kDa isolated from the CF of *Dendrobaena veneta*, activates human macrophages. This fraction, now called Venetin-1, caused them to secrete some of the most important cytokines, involved in many immune processes in the human body [38]. These findings suggest that Venetin-1 could potentially be used in non-specific immunotherapy.

To verify Venetin-1's safety and efficacy in stimulating an immune response, cyclophosphamide (CP) was used to induce immunosuppression prior to Venetin-1 treatment. These are the major steps of the experiment:

A. Treatment of mice with CP and Venetin-1
B. Euthanasia and spleen isolation
C. Spleen tissue homogenization and cell lysis
D. FASP protein digestion and peptide purification
E. Quantitative SWATH-MS spectrum registration
F. Statistical and functional protein analyses

6. MATERIALS AND METHODS

6.1 Mice and reagents

The study involved 12 mice of the C57BL/6 strain (7-8 week old, 18–22 g weight). The Local Ethical Committee for Animal Experiments, University of Life Sciences in Poznań, approved the study (Approval No. 5/2022, 25.02.2022). Mice were single-housed under a 12/12 h light/dark cycle. They were kept at room temperature (21 ± 2 °C) with a relative humidity of 55–60%. Animals were fed with water and a standard laboratory diet.

PhD Marta Fiołka provided the protein-sugar fraction Venetin-1 from the Immunobiology Department, Faculty of Biology and Biotechnology at UMCS University in Lublin. Coelomic fluid was collected from the body cavity of *Dendrobaena veneta* earthworms by the electric shock method at the voltage of 4,5 V. Isolated fluid was centrifuged for 10 minutes at 5,000 rpm to remove coelomocytes. After it was filtered through Millipore strainers with a pore diameter of 0.22 µm, sterilized fluid was heated at 70°C for 10 minutes and dialyzed in water in a dialysis bag with a pore diameter that cuts off compounds of more than 14 kDa. Dialysis was carried out for 24 hours at 4°C. The resulting fraction of compounds with a mass of more than 14 kDa was tested for protein and non-protein components, and after their separation, a protein-sugar fraction Venetin-1 was obtained. Fluid was stored at -20°C before usage. [36].

6.2 Protocols for CP and Venetin-1 treatment

As part of the study of the effects of Venetin-1 on the organs of mice, 12 individuals were selected and divided into four groups, with three in each (Fig. 2). Every group was administered treatment by intraperitoneal injection. The first group (later called 'NaCl') was an intact control. This group received 0.9% NaCl solution for 14 days. The second group was a cyclophosphamide control (called 'CP'). During the first three days, mice were treated with CP at 80 mg/kg of body weight. Afterwards, they received NaCl for seven days. The third and fourth were the Venetin-1 study groups. They were also administered to CP for the first three days at the mentioned concentration. Later, mice from the third and fourth groups were treated with Venetin-1 at 50 mg/kg b. w. (called 'CP+50V') or 100 mg/kg b. w. (called 'CP+100V'), respectively.

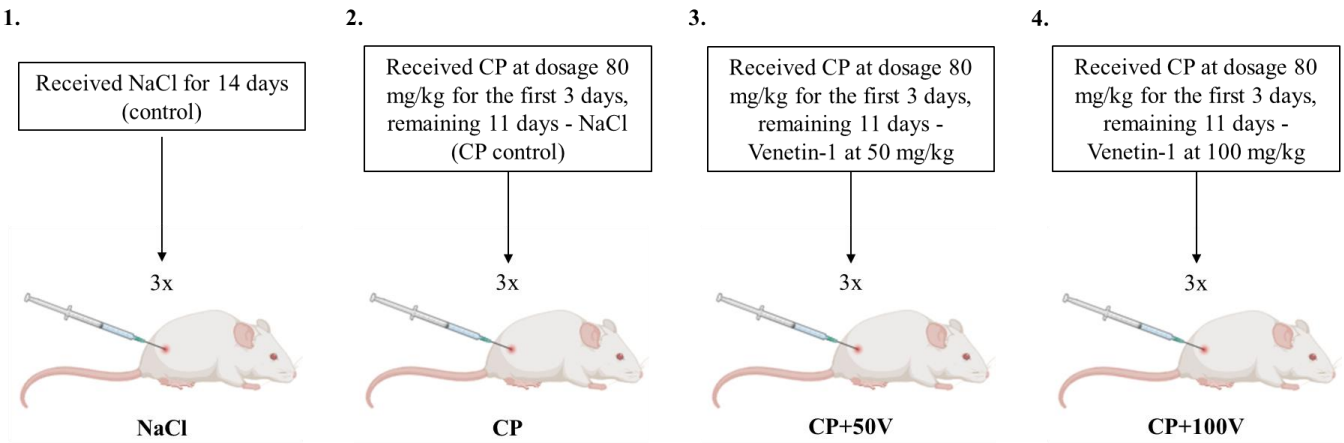

Figure 2. Scheme presenting different mice treatment protocols. CP - Cyclophosphamide.

6.3 Homogenization of the spleen and cell lysis

The spleen isolated from the mouse was washed several times with phosphate buffer (10 mM) to remove blood particles. After that, homogenisation was performed using a manual tissue homogeniser (Omni TH with Omni Tip, Omni International). The process was carried out in a lysis buffer containing 1% sodium dodecyl sulfate (SDS), dithiothreitol and 100 mM TRIS-HCl. Then, the samples with spleen cells were sonicated (QSonica).

6.4 Protein digestion using FASP procedure

Protein digestion of the spleen lysate was performed using the Filter-Aided Sample Preparation (FASP) method (Fig. 3). There are four critical steps in this procedure: removal of unfavourable low molecular weight components in a buffer containing urea, carbamidomethylation of thiols, protein digestion and elution of peptides [39]. Thus, samples with spleen lysate around 100 µg were mixed with 200µl of 8M urea solution in 100 mM Tris-HCl pH 8,5 (UA). Mixtures were applied on Microcon® (Merck Millipore) centrifugal filters with a molecular weight cutoff of 10 kDa. Microcon® filters with samples were centrifuged at 10000 g for 20 minutes. Afterwards, filters were washed with 200 µl of UA and centrifuged at the same conditions. The washing step was repeated three times. To alkylate the proteins, 100 µl of IAA (55 mM iodoacetamide in UA) was added to the filters, which were incubated for 20 minutes in the dark. After the incubation, filters were centrifuged and washed with UA buffer three times. The next step was washing filters with DB buffer (50 mM Tris-HCl pH 8,5). This step was repeated twice. Washed filters were treated with 2 µg of Trypsin enzyme in DB buffer overnight at 37 ˚C. The next day, elution of the peptides was performed. Filters after protein digestion were centrifuged and washed twice with 125 µl and 100 µl of DB buffer, respectively.

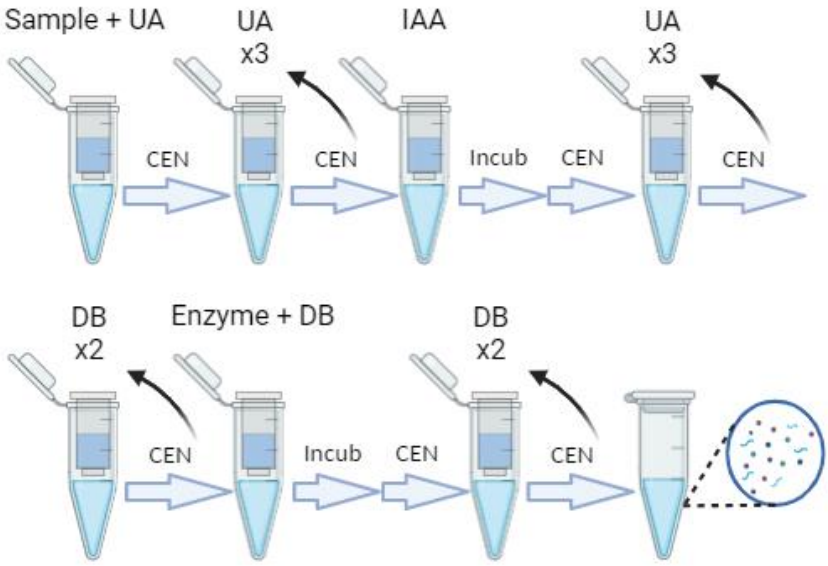

Figure 3. Protein digestion scheme using Filter-Aided Sample Preparation (FASP) method. CEN – centrifugation; Incub – Incubation.

6.4 Peptide Purification

Eluted peptides were purified and desalted using the StageTips method with pipet tips containing the C18 solid phase. Every time, the flowthroughs of liquids were forced by centrifuging. Firstly, methanol was added to moisten the solid phase, after which tips were washed with 60 % acetonitrile (ACN) and 1% acetic acid (AA) solution to remove contaminants. After the washing, the sorbent was conditioned with 1% AA and the 10 µg of peptides diluted in 100 µl of 1% AA was added. The next step was the elution of peptides captured in the solid phase. Tips were put into the tubes, and 60% ACN in 1% AA solution was added to the sorbent. Samples were condensed using a SpeedVac concentrator [40].

6.5 Mass spectrometry analysis LC/MS-MS – SWATH-MS

Spectrum registration was performed on a TripleTOF 5600+ (Sciex Framingham, MAmass spectrometer, which was connected to a liquid chromatography system, the Ekspert MicroLC 200 Plus System (Eksigent, Redwood City, CA). Each sample was analyzed in three technical repeats. Sampled were applied into the LC column using autosampler CTC Pal (CTC Analytics AG, Zwinger, Switzerland) performing 5 µl injections. Buffer A consisted of 0,1% formic acid (FA) in water, while buffer B consisted of 0,1% FA in ACN. Chromatographic separations were performed on the ChromXP C18CL column (3 µm, 120 Å, 150 × 0.3 mm). The chromatographic gradient was based on the mentioned A and B buffers. The B buffer gradient increased from 11% to 42% in 60 minutes. The Analyst TF 1.7.1 software (SCIEX) controlled the whole system. First measurements of all 12 spleen samples served for library construction. Measurements for the spectral library were acquired in data-dependent acquisition (DDA) mode. SWATH method with variable windows was prepared based on the SWATHTuner5 method. Fragmentation was performed in 25 variable-width windows in high sensitivity mode with ion spectra accumulation in 50 ms. Registered Spectrum was uploaded to the PeakView program to choose proteins with at least two identified peptides.

6.6 Statistical and functional analyses

Statistical significance and fold change (FC) calculations were performed in the Perseus program [41]. Firstly, samples were divided into groups by biological and then technical repeats. Afterwards, the correlation between samples was measured (Pearson Correlation). Statistical T-Test was performed with statistical significance at q-value < 0,05. For the FC measurements, groups 2, 3 and 4 described in the 6.2 paragraph were compared to the 'NaCl' control group. In other FC calculations, the groups 3 and 4 were compared to the 'CP' control group. Functional analysis was conducted in STRING 11.0 for the interaction visualization visualisation and KEGG pathway identification for up- and down-regulated proteins.

7. RESULTS

7.1 Effects of animal treatment

In the planned study, the immunomodulatory properties of Venetin-1 were investigated in immunosuppressed mice. Immunodeficiency was induced via cyclophosphamide (CP) treatment followed by Venetin-1 injection at 50 mg/kg or 100 mg/kg. The following steps of the study were - isolation of spleen, homogenisation, protein digestion using the FASP (Filter Aided Sample Preparation) method, peptide purification, spectrum registration SWATH-MS (Sequential Window Acquisition of all Theoretical Mass Spectra), statistical and function analysis of the proteins. The procedure was described earlier in Materials and Methods per the following scheme (Fig. 4).

To mark, during the experiment, mice exhibited no signs of a negative impact on their vital functions, and none of them died before the study concluded. However, all mice injected with CP displayed lethargy, dull pelage, fur piloerection, and reduced food and water intake. After the termination of CP injections, mice slowly recovered.

7.2 Statistical analysis of SWATH-MS data

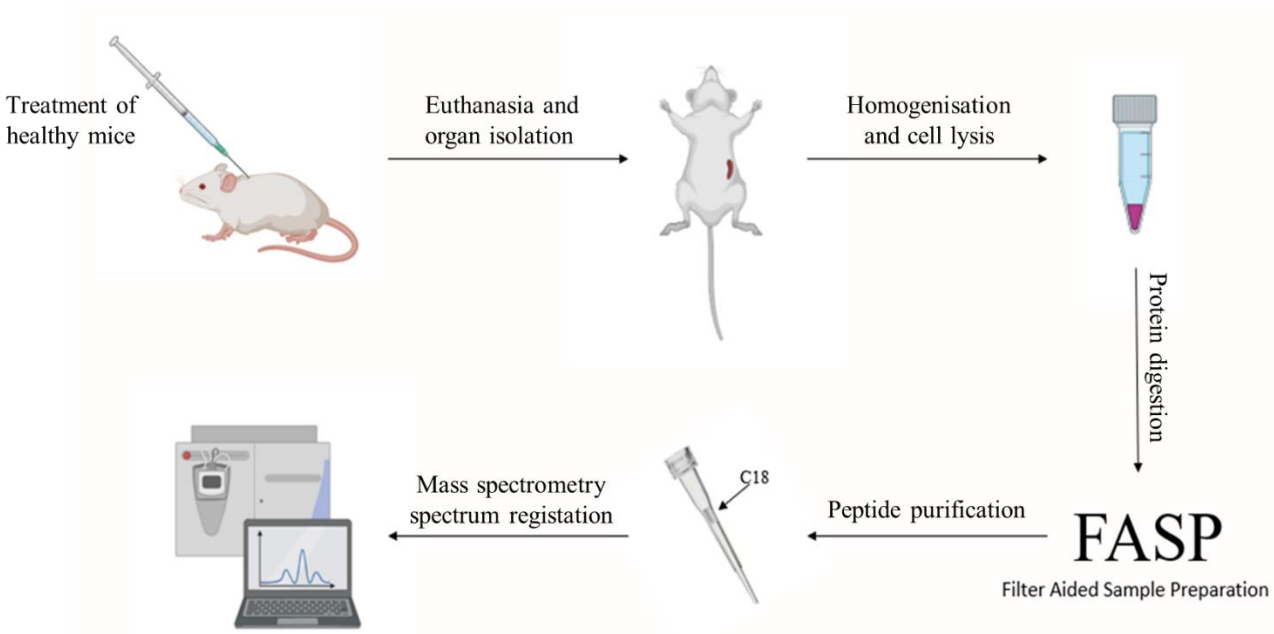

Figure 4. The scheme presenting stages of research following Cyclophosphamide and Venetin-1 treatment of mice.

The quantitative analysis of spleen tissue samples was performed using the Sequential Window Acquisition of all Theoretical mass spectra (SWATH-MS) methodology. Registered Spectrum was uploaded to the PeakView program to review identified proteins. While in Perseus the statistical significance and fold change were calculated as described in paragraph 6.6 Statistical and functional analysis. Groups ‘CP’, ‘CP+50V’ and ‘CP+100V’ were compared to the control group ‘NaCl’. Moreover, groups ‘CP+50V’ and ‘CP+100V’ were compared to the ‘CP’ control. Proteins were considered significant if they met the following criteria: $q < 0{,}05$ and Fold Change (FC) > 2 or < 0,55. Table 3 displays the total number of up-and-down-regulated proteins in each analysed group compared to the control ones. In accordance with the

results presented in the table, the 3-day cyclophosphamide treatment ('CP' group) caused significant changes in 36 proteins. After Venetin-1 injection at 50 mg/kg b. w. to immunosuppressed mice ('CP+50V'), changes in protein levels were minimal. On the other hand, a higher dose of Venetin-1 ('CP+100V') resulted in even more significant protein changes than CP alone. However, both of the Venetin-1 groups showed the most important difference to the CP control.

Table 3. The number of proteins whose level was significantly changed in mouse spleen tissue after the cyclophosphamide and Venetin-1 treatment compared to the control group. 'NaCl' – a group of mice treated only with 0,9% NaCl; 'CP' – a group of mice treated with cyclophosphamide (80 mg/kg) and NaCl; 'CP+50V' – mice treated with cyclophosphamide and Venetin-1 (50 mg/kg); 'CP+100V' – mice treated with cyclophosphamide and Venetin-1 (100 mg/kg). q < 0,05.

Study group / control group	CP / NaCl	CP+50V / NaCl	CP+100V / NaCl	CP+50V / CP	CP+100V / CP
Number of proteins	36	1	201	21	221

7.3 KEGG pathways enrichment analysis

To correlate the quantitative analysis results with protein functions, proteins were categorized as up-regulated or down-regulated. The data were analyzed using the STRING software [42], where a KEGG pathway enrichment analysis was conducted [43]. Table 4 lists the KEGG pathways identified among the up-regulated proteins, ordered by ascending false discovery rate (FDR). In the 'CP' group, only one KEGG pathway, 'African trypanosomiasis', was identified. The 'Cp+100V' group exhibited up-regulation of 23 pathways compared to the 'NaCl' control and changes in 38 pathways versus 'CP' group. Some with the low FDR in both comparisons were 'Regulation of actin cytoskeleton', 'Tight junction', 'Antigen processing and presentation', 'Bacterial invasion of epithelial cells', 'Platelet activation' etc.

Table 4. KEGG pathways identified in up-regulated proteins in study groups compared to control groups. 'NaCl' – group of mice treated only with 0,9% NaCl; 'CP' – group of mice treated with cyclophosphamide (80 mg/kg) and NaCl; 'CP+100V' – mice treated with cyclophosphamide and Venetin-1 (100 mg/kg); FDR – false discovery rate. $q < 0,05$; FC (fold change) > 2.

Study group / control group	**KEGG Pathway**	**Description**	**Observed gene count**	**Background gene count**	**FDR**
CP / NaCl	mmu05143	African trypanosomiasis	3	35	0.0024
CP+100V / NaCl	mmu04810	Regulation of actin cytoskeleton	11	211	1.98e-08
	mmu04141	Protein processing in endoplasmic reticulum	8	169	9.91e-06
	mmu04612	Antigen processing and presentation	6	75	1.78e-05
	mmu04510	Focal adhesion	7	196	0.00018
	mmu04611	Platelet activation	6	120	0.00018
	mmu05205	Proteoglycans in cancer	7	196	0.00018
	mmu04530	Tight junction	6	152	0.00039
	mmu04270	Vascular smooth muscle contraction	5	133	0.0025
	mmu04261	Adrenergic signalling in cardiomyocytes	5	141	0.0029
	mmu05100	Bacterial invasion of epithelial cells	4	71	0.0029
	mmu04657	IL-17 signalling pathway	4	89	0.0054
	mmu04670	Leukocyte transendothelial migration	4	113	0.0119
	mmu05132	Salmonella infection	5	212	0.0127
	mmu03010	Ribosome	4	126	0.0152

	mmu04915	Estrogen signalling pathway	4	131	0.0164
	mmu05418	Fluid shear stress and atherosclerosis	4	142	0.0206
	mmu04971	Gastric acid secretion	3	69	0.0262
	mmu05133	Pertussis	3	73	0.0274
	mmu05206	MicroRNAs in cancer	4	159	0.0274
	mmu04260	Cardiac muscle contraction	3	84	0.0384
	mmu05410	Hypertrophic cardiomyopathy	3	88	0.0415
	mmu05414	Dilated cardiomyopathy	3	90	0.0422
	mmu04966	Collecting duct acid secretion	2	27	0.0497
CP+100V / CP	mmu01200	Carbon metabolism	9	122	8.72e-07
	mmu01230	Biosynthesis of amino acids	8	77	8.72e-07
	mmu04810	Regulation of actin cytoskeleton	11	211	8.72e-07
	mmu04530	Tight junction	7	152	0.00037
	mmu05132	Salmonella infection	8	212	0.00037
	mmu00010	Glycolysis / Gluconeogenesis	5	65	0.00065
	mmu04670	Leukocyte transendothelial migration	6	113	0.00065
	mmu05100	Bacterial invasion of epithelial cells	5	71	0.00074
	mmu01100	Metabolic pathways	19	1534	0.00086
	mmu04510	Focal adhesion	7	196	0.00086
	mmu05205	Proteoglycans in cancer	7	196	0.00086
	mmu05418	Fluid shear stress and atherosclerosis	6	142	0.0011
	mmu05146	Amoebiasis	5	102	0.0024
	mmu04151	PI3K-Akt signalling pathway	8	353	0.0036
	mmu04611	Platelet activation	5	120	0.0043
	mmu04971	Gastric acid secretion	4	69	0.0055
	mmu05133	Pertussis	4	73	0.0063
	mmu04612	Antigen processing and presentation	4	75	0.0066
	mmu04261	Adrenergic signalling in cardiomyocytes	5	141	0.0069
	mmu00020	Citrate cycle (TCA cycle)	3	32	0.0073
	mmu00030	Pentose phosphate pathway	3	33	0.0073
	mmu05200	Pathways in cancer	9	527	0.0073
	mmu04260	Cardiac muscle contraction	4	84	0.0078
	mmu04657	IL-17 signalling pathway	4	89	0.0088
	mmu05410	Hypertrophic cardiomyopathy	4	88	0.0088
	mmu05414	Dilated cardiomyopathy	4	90	0.0088
	mmu04066	HIF-1 signalling pathway	4	110	0.0174
	mmu04114	Oocyte meiosis	4	114	0.0189
	mmu04722	Neurotrophin signalling pathway	4	114	0.0189
	mmu05203	Viral carcinogenesis	5	194	0.0189
	mmu04915	Estrogen signalling pathway	4	131	0.0281
	mmu04270	Vascular smooth muscle contraction	4	133	0.0287
	mmu04520	Adherens junction	3	68	0.0345
	mmu01210	2-Oxocarboxylic acid metabolism	2	20	0.0365
	mmu04921	Oxytocin signalling pathway	4	145	0.0365
	mmu05012	Parkinson disease	5	238	0.0365
	mmu04390	Hippo signalling pathway	4	155	0.0423
	mmu05014	Amyotrophic lateral sclerosis	6	362	0.0423

Table 5 displays the KEGG-enriched pathways for the down-regulated proteins. Only one pathway was identified in the 'CP' group, called 'Ribosome'. This pathway also appeared in the 'CP+100V' group with the lowest FDR rate. In total, in the 'CP+100V' group, 32 KEGG pathways were recognised versus the 'NaCl' control and 21 when compared to the 'CP' control. Besides the 'Ribosome', other pathways with the notably low FDR rate were associated with Parkinson's, Huntington's, Prion and Alzheimer's diseases and the different metabolic pathways.

Table 5. KEGG pathways were identified in down-regulated proteins in study groups compared to control groups. 'NaCl' – group of mice treated only with 0,9% NaCl; 'CP' – group of mice treated with cyclophosphamide (80 mg/kg) and NaCl; 'CP+100V' – mice treated with cyclophosphamide and Venetin-1 (100 mg/kg); / – versus; FDR – false discovery rate. $q < 0,05$; FC < 0,55.

Study group / control group	KEGG Pathway	Description	Observed gene count	Background gene count	FDR
CP / NaCl	mmu03010	Ribosome	4	126	0.00027
CP+100V / NaCl	mmu03010	Ribosome	22	211	4.43e-22
	mmu05012	Parkinson disease	18	238	3.43e-12
	mmu01100	Metabolic pathways	38	1534	7.33e-12
	mmu05016	Huntington disease	18	297	6.00e-11
	mmu05020	Prion disease	17	263	8.25e-11
	mmu05010	Alzheimer disease	18	360	8.44e-10
	mmu01200	Carbon metabolism	10	122	3.71e-07
	mmu05014	Amyotrophic lateral sclerosis	15	362	4.00e-07
	mmu00280	Valine, leucine and isoleucine degradation	7	55	3.49e-06
	mmu00190	Oxidative phosphorylation	9	130	5.81e-06
	mmu00071	Fatty acid degradation	6	50	3.36e-05
	mmu04714	Thermogenesis	10	225	4.79e-05
	mmu01212	Fatty acid metabolism	6	61	8.27e-05
	mmu00640	Propanoate metabolism	5	34	8.72e-05
	mmu00650	Butanoate metabolism	4	28	0.00090
	mmu00020	Citrate cycle (TCA cycle)	4	32	0.0013
	mmu01230	Biosynthesis of amino acids	5	77	0.0028
	mmu04216	Ferroptosis	4	40	0.0028
	mmu03050	Proteasome	4	46	0.0042
	mmu05017	Spinocerebellar ataxia	6	140	0.0044
	mmu00380	Tryptophan metabolism	4	51	0.0054
	mmu01210	2-Oxocarboxylic acid metabolism	3	20	0.0054
	mmu04932	Non-alcoholic fatty liver disease	6	147	0.0054
	mmu04217	Necroptosis	6	153	0.0058
	mmu00310	Lysine degradation	4	61	0.0087
	mmu00062	Fatty acid elongation	3	29	0.0121
	mmu00480	Glutathione metabolism	4	71	0.0138
	mmu03040	Spliceosome	5	131	0.0171
	mmu04260	Cardiac muscle contraction	4	84	0.0231
	mmu04146	Peroxisome	4	85	0.0233
	mmu05322	Systemic lupus erythematosus	4	93	0.0308
	mmu04145	Phagosome	5	162	0.0365
CP+100V / CP	mmu03010	Ribosome	23	126	3.90e-24
	mmu05012	Parkinson disease	18	238	1.31e-12
	mmu05020	Prion disease	18	263	4.38e-12

mmu05016	Huntington disease	18	297	2.33e-11
mmu05010	Alzheimer disease	18	360	4.00e-10
mmu05014	Amyotrophic lateral sclerosis	14	362	1.96e-06
mmu00190	Oxidative phosphorylation	9	130	5.23e-06
mmu01100	Metabolic pathways	27	1534	7.91e-06
mmu04216	Ferroptosis	5	40	0.00022
mmu04714	Thermogenesis	9	225	0.00028
mmu01210	2-Oxocarboxylic acid metabolism	4	20	0.00030
mmu01230	Biosynthesis of amino acids	5	77	0.0030
mmu01200	Carbon metabolism	5	122	0.0213
mmu00020	Citrate cycle (TCA cycle)	3	32	0.0250
mmu05017	Spinocerebellar ataxia	5	140	0.0333
mmu04260	Cardiac muscle contraction	4	84	0.0344
mmu04932	Non-alcoholic fatty liver disease	5	147	0.0362
mmu00860	Porphyrin and chlorophyll metabolism	3	41	0.0379
mmu04217	Necroptosis	5	153	0.0383
mmu03050	Proteasome	3	46	0.0463
mmu04145	Phagosome	5	162	0.0463

7.4 Common altered proteins in Venetin-1 treatment groups

To better understand Venetin-1's effects, further analysis focused on proteins exclusively altered in groups treated with Venetin-1. Thus, Venn diagrams (Fig. 5) helped to illustrate unique and familiar up-regulated and down-regulated proteins across all groups.

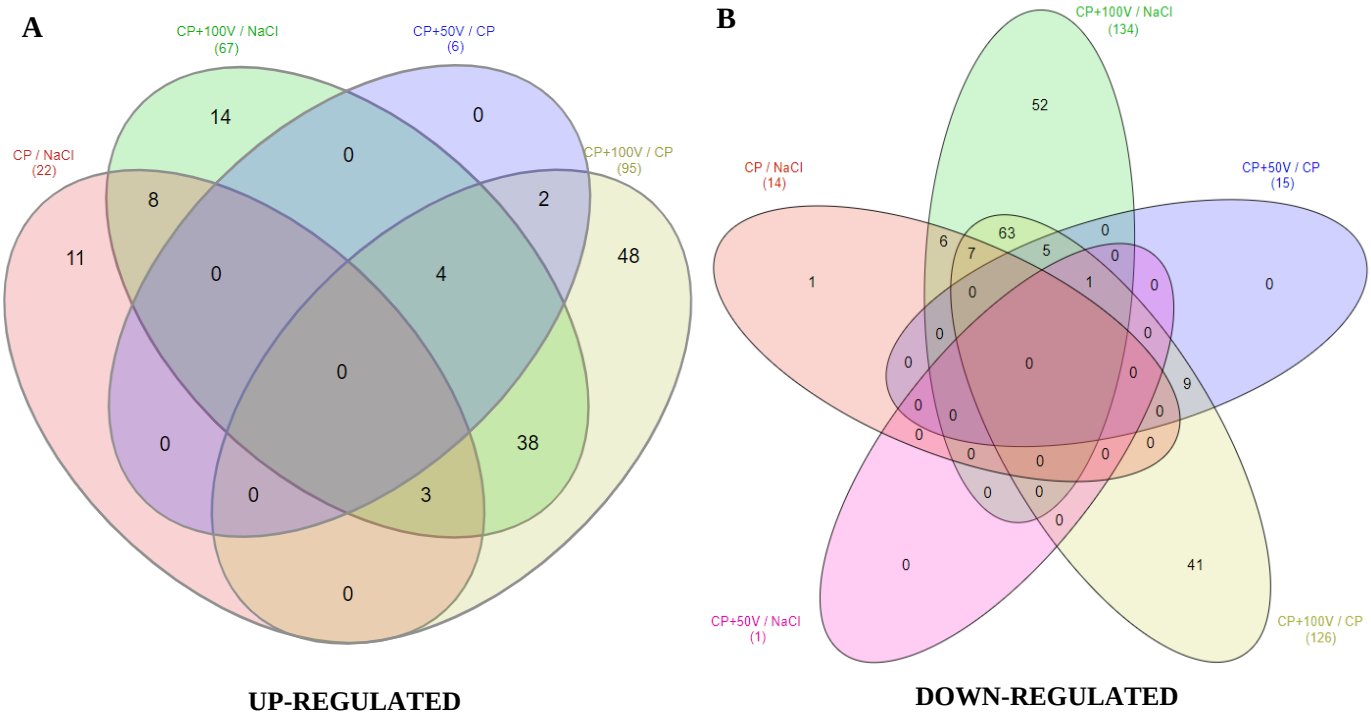

Figure 5. Interactive Venn diagrams of up-regulated [A] and down-regulated [B] proteins. 'NaCl' – group of mice treated only with 0,9% NaCl; 'CP' – group of mice treated with cyclophosphamide (80 mg/kg) and NaCl; 'CP+50V' – mice treated with cyclophosphamide and Venetin-1 (50 mg/kg); 'CP+100V' – mice treated with cyclophosphamide and Venetin-1 (100 mg/kg).

Comparison ‘CP+50V / NaCl’ is absent in illustration [A] of Venn diagrams (Fig. 5) because it lacked up-regulated proteins. Moreover, ‘CP+50V’ was the only group with no unique proteins, which exhibited up- or down-regulation.

Proteins that were common between the 'CP+50V' and 'CP+100V' groups or present in both control comparisons within one of the Venetin-1 groups were identified. According to the Venn diagram, comparisons ‘CP+100V / NaCl’, ‘CP+50V / CP’ and ‘CP+100V / CP’ share four up-regulated proteins. All of them are displayed in Table 6, along with their fold change (FC) and information found in other publications. Additionally, two proteins were shared between ‘CP+50V / CP’ and ‘CP+100V / CP’. Moreover, up to 38 common proteins appeared in both control comparisons of ‘CP+100V’. Ones that featured high FC value are also displayed.

Table 6. Examples of significantly up-regulated proteins in Venetin-1 treatment groups and relevant information from various publications. ‘NaCl’ – group of mice treated only with 0,9% NaCl; ‘CP’ – group of mice treated with cyclophosphamide (80 mg/kg) and NaCl; ‘CP+50V’ – mice treated with cyclophosphamide and Venetin-1 (50 mg/kg); ‘CP+100V’ – mice treated with cyclophosphamide and Venetin-1 (100 mg/kg). $q < 0,05$; $FC > 2$. HSP90 - heat shock protein HSP 90; CKAP4 - cytoskeleton-associated protein 4; PLEK – pleckstrin; Myh9 - myosin-9; MANF - mesencephalic astrocyte-derived neurotrophic factor.

Study group / control group	FC	Protein	Quote from publication	Reference
CP+100V / NaCl CP+50V / CP CP+100V / CP	3,67 2,05 2,92	Fibronectin	„Fibronectin 1 promotes melanoma proliferation and metastasis by inhibiting apoptosis”	[44]
CP+100V / NaCl CP+50V / CP CP+100V / CP	12,09 3,75 5,7	Heat shock protein HSP 90-alpha	„Overexpression of HSP90 has been linked …, viral infection, inflammation”	[45]
CP+100V / NaCl CP+50V / CP CP+100V / CP	13,60 11,51 17,36	Fibrinogen beta chain	„Increased levels of fibrinogen have been found in many inflammatory diseases”	[46]
CP+100V / NaCl CP+50V / CP CP+100V / CP	2,87 4,23 5,33	Cytoskeleton-associated protein 4	„ CKAP4 is a receptor for DKK1 and DKK1/CKAP4 signalling promotes normal and cancer cell proliferation”	[47]
CP+100V / NaCl CP+100V / CP	14,23 12,45	S100A9 protein	„S100A8, S100A9 and S100A12 play a crucial role in resisting microbial infection and maintaining immune homeostasis”	[48]
CP+100V / NaCl CP+100V / CP	19,74 9,68	S100A8 protein		
CP+100V / NaCl CP+100V / CP	50,58 11,09	Pleckstrin	„PLEK is strongly positively correlated with most lymphocytes”	[49]
CP+100V / NaCl CP+100V / CP	39,68 6,88	Myosin-9	„Fundamental role of Myh9 for neutrophil trafficking in innate immunity”	[50]
CP+100V / NaCl CP+100V / CP	30,28 5,18	Mesencephalic astrocyte-derived neurotrophic factor	„(MANF) that biases immune cells toward an anti-inflammatory phenotype”	[51]

In the case of down-regulated proteins, six proteins were shared between the ‘CP+50V’ and ‘CP+100V’ groups. One of which, Poly(rC)-binding protein 1, even appeared in every control comparison. All of them, along with FC value and quotes from articles, are listed in Table 7. Also, nine common proteins were identified in comparisons ‘CP+50V / CP’ and ‘CP+100V / CP’. Finally, both comparisons of the ‘CP+100V’ group shared 63 down-regulated proteins, some of which are also presented.

Table 7. Examples of significantly down-regulated proteins in Venetin-1 treatment groups and relevant information from various publications. 'NaCl' – a group of mice treated only with 0,9% NaCl; 'CP' – a group of mice treated with cyclophosphamide (80 mg/kg) and NaCl; 'CP+50V' – mice treated with cyclophosphamide and Venetin-1 (50 mg/kg); 'CP+100V' – mice treated with cyclophosphamide and Venetin-1 (100 mg/kg). q < 0,05; FC < 0,55. PCBP-1 - poly(rC)-binding protein 1; E-Syt1 - extended synaptotagmin-1; HSC - haematopoietic stem cells; CLTC - clathrin heavy chain; SERPINA3K - serine protease inhibitor A3K; NK - natural killer; DDX21 - nucleolar RNA helicase 2; IFN-I - type-I interferon.

Study group / control group	FC	Protein	Quote from publication	Reference
CP+50V / NaCl CP+100V / NaCl CP+50V / CP CP+100V / CP	0,11 0,13 0,18 0,22	Poly(rC)-binding protein 1	„PCBP-1 has important roles in cellular Iron homeostasis, mitochondrial stability, and other cellular activities involved in …. neurodegenerative diseases"	[52]
CP+100V / NaCl CP+50V / CP CP+100V / CP	0,11 0,15 0,08	Tubulin beta-4B chain	„Apicularen A also disrupted the formation of the normal microtubule architecture by down-regulating tubulin synthesis and thus might inhibit tumour colonization to the liver"	[53]
CP+100V / NaCl CP+50V / CP CP+100V / CP	0,25 0,13 0,17	Tubulin alpha-4A chain		
CP+100V / NaCl CP+50V / CP CP+100V / CP	0,52 0,31 0,27	Extended synaptotagmin-1	„E-Syts downmodulate T-cell receptor signal ing, T-cell-mediated cytotoxicity, degranulation, and cytokine production"	[54]
CP+100V / NaCl CP+50V / CP CP+100V / CP	0,23 0,11 0,12	Histone H3.3	„H3.3, but not its chaperone Hira, prevents premature HSC exhaustion and differentiation into granulocyte-macrophage progenitors"	[55]
CP+100V / NaCl CP+50V / CP CP+100V / CP	0,25 0,13 0,17	Ferritin light chain 1	„Ferritin light chain has been shown to reduce NF-κB activation in Raw 264.7 macrophages"	[56]
CP+100V / NaCl CP+100V / CP	0,15 0,10	Clathrin heavy chain 1	„ Down-regulation of CLTC resulted in tumour-suppressive effects in vitro and in vivo"	[57]
CP+100V / NaCl CP+100V / CP	0,15 0,14	Serine protease inhibitor A3K	„ SERPINA3K down-regulated the expression of the pro-angiogenic and pro-inflammatory factors"	[58]
CP+100V / NaCl CP+100V / CP	0,09 0,08	Nucleolar RNA helicase 2	„knockout of DDX21 enhanced Sendai virus (SeV)-induced IFN-β production"	[59]
CP+100V / NaCl CP+100V / CP	0,30 0,41	Probable ATP-dependent RNA helicase DDX5	„DDX5 knockdown significantly promoted DNA or RNA virus infection-induced IFN-I production"	[60]

8. DISCUSSION

Cancer is a range of diseases characterised by abnormal and uncontrolled cell growth, which causes millions of deaths every year. Extensive research has been and continues to be conducted in this field, leading to a much deeper understanding of the molecular pathways involved in disease progression. This growing knowledge also brought new medicine, including targeted therapies and immunotherapies. These are highly specific and very promising treatments; however, they are not suitable for every case of tumour or patient. Therefore, chemotherapy continues to be a widely chosen option despite its life-threatening side effects. What is more, chemotherapeutic agents have remained a primary choice in advanced-stage cancers, especially in cases where surgery or radiation therapy are not suitable due to various constraints [61]. Thus, it is still essential to keep improving chemotherapy by minimising its side effects, such as immunosuppression.

This project aimed to investigate the immunomodulatory properties and safety of Venetin-1 by performing its first *in vivo* study on mice. Before Venetin-1's administration, a widespread chemotherapy agent, cyclophosphamide, was used to induce immunodeficiency in tested animals. That way, Venetin-1's ability to restore immune response was examined. The idea of this project comes from previous *in vitro* studies, where Venetin-1 demonstrated its anti-tumor activity [34], [35] and ability to induce secretion of crucial pro-inflammatory cytokines [38]. Moreover, it did not show cytotoxicity against normal vertebrate cells.

The study began with randomly selecting 12 healthy mice, which were later divided into 4 groups. The first was administered with NaCl (control), and the second with cyclophosphamide and NaCl (CP control). The remaining two groups were treated with CP and Venetin-1 at two different concentrations – 50 mg/kg and 100 mg/kg. After 14 days of treatment, mice were euthanised, and selected organs were isolated. In this project, the focus was on spleen tissue examination. The critical steps of sample preparation were – cell lysis, FASP protein digestion, peptide purification and mass spectrometry analysis.

The statistical analysis of up- and down-regulated proteins showed that cyclophosphamide induced significant changes in the levels of 36 proteins compared to the NaCl control group. In contrast, administering a lower dose of Venetin-1 (50 mg/kg) to immunosuppressed mice changed the level of only one protein, suggesting that Venetin-1 at this dosage effectively normalised the protein levels altered by CP. However, a higher dose of Venetin-1 (100 mg/kg) led to significant changes in 201 protein levels, indicating that beyond a specific concentration, Venetin-1 causes a severe impact on protein profile.

The KEGG pathways in the STRING database were identified during the functional analysis. Within the up-regulated proteins, only one pathway was recognised in the 'CP' group, called 'African trypanosomiasis'. This pathway is known to be causing immunodeficiency [62], which aligns with the expected immunosuppressive effects of CP treatment. Numerous up-regulated KEGG pathways were identified in the '100V+CP' group, many involved in immune processes such as antigen processing and presentation, platelet activation, and the IL-17 signalling pathway. Additionally, many pathways were related to cellular proliferation and dynamics. Notable examples include estrogen, PI3K-Akt, and Hippo signalling pathways and pathways involved in regulating the actin cytoskeleton, tight junctions, biosynthesis of amino acids, and protein processing. Similar findings emerged from a search about proteins

exclusively upregulated in groups treated with Venetin-1. Interestingly, two proteins, S100A, appeared significantly up-regulated in every control comparison of the ‘CP+100V’ group. S100A proteins are known to initiate inflammatory signal transduction, induce cytokine expression, and play a role in immune regulation. Additionally, they are involved in regulating cell proliferation and tumorigenesis [48].

In the context of down-regulation, in the ‘CP’ group, one KEGG pathway was recognized, called a ribosome. This pathway, having the lowest false discovery rate (FDR), was also observed in the 'CP+100V' group. Interestingly, it appeared among the up-regulated KEGG pathways in ‘CP+100V’ compared to ‘NaCl’ control. The consequences of up-or down-regulation of the ribosome pathway remain unclear. It likely depends on which specific proteins are affected. Additionally, the lowest FDR pathways in the ‘CP+100V’ group were linked to neurodegenerative diseases. These are closely associated with oxidative stress and mitochondrial dysfunction [63]. The down-regulation of these pathways could indicate that Venetin-1 has an antioxidative effect. In contrast, a previous in vitro study of Venetin-1’s impact on the A549 cancer cell line showed up-regulation of similar neurodegenerative pathways [34]. This difference is likely due to Venetin-1’s cytotoxic effect on A549 cells.

To summarise, during performed studies, Venetin-1 confirmed its immunomodulatory properties by up-regulating immune activation proteins, suggesting its potential application in non-specific immunotherapy. This finding also makes it a promising ingredient in vaccines, drugs, and treatments for certain cancers, infectious diseases, and other immunocompromised conditions. In future research, to investigate Venetin-1’s possible use in immunotherapy, it should be tested *in vivo* on cancer models. Also, the optimal concentration of Venetin-1, which balances efficacy and safety, should be obtained. Moreover, further investigation is needed to understand its mechanism of action better.

9. REFERENCES

1. V.T. DeVita, E. Chu; A History of Cancer Chemotherapy. *Cancer Res*, 2008. https://doi.org/10.1158/0008-5472.CAN-07-6611
2. F.C. Meotti, S. Forner, J.F. Lima-Garcia, A.F. Viana, J.B. Calixto; Antagonism of the transient receptor potential ankyrin 1 (TRPA1) attenuates hyperalgesia and urinary bladder overactivity in cyclophosphamide-induced haemorrhagic cystitis. *Chem Biol Interact*, 2013. https://doi.org/10.1016/j.cbi.2013.03.008
3. M.J. Moore; Clinical pharmacokinetics of cyclophosphamide. *Clin Pharmacokinet*, 1991. https://doi.org/10.2165/00003088-199120030-00002
4. M. Ahlmann, G. Hempel; The effect of cyclophosphamide on the immune system: implications for clinical cancer therapy. *Cancer Chemother Pharmacol*, 2016. https://doi.org/10.1007/s00280-016-3152-1
5. R.A. Fleming; An overview of cyclophosphamide and ifosfamide pharmacology. *Pharmacotherapy*, 1997. PMID: 9322882
6. E. Şengü, V. Gelen, S. Gedikli, S. Özkanlar, C. Gür, F. Çelebi, A. Çınar; The protective effect of quercetin on cyclophosphamide-Induced lung toxicity in rats. *Biomedicine and Pharmacotherapy*, 2017. https://doi.org/10.1016/j.biopha.2017.05.047
7. L.H. Fraiser, S. Kanekal, J.P. Kehrer. Cyclophosphamide toxicity. Characterising and avoiding the problem. *Drugs*, 1991 Nov. 10.2165/00003495-199142050-00005
8. M.A. Ayza, K.A. Zewdie, B.A. Tesfaye, D.Z. Wondafrash, A.H. Berhe; The Role of Antioxidants in Ameliorating Cyclophosphamide-Induced Cardiotoxicity. *Oxid Med Cell Longev*, 2020. https://doi.org/10.1155/2020/4965171
9. R. Jeelani, S.N. Khan, F. Shaeib, H.-Re. Kohan-Ghadr, S.R. Aldhaheri, T. Najafi, M. Thakur, R. Morris, H. M. Abu-Soud; Cyclophosphamide and acrolein induced oxidative stress leading to deterioration of metaphase II mouse oocyte quality. *Free Radic. Biol. Med.* 2017. https://doi.org/10.1016/j.freeradbiomed.2017.05.006
10. K.A. Mills, R. Chess-Williams, C. McDermott; Novel insights into the mechanism of cyclophosphamide-induced bladder toxicity: chloroacetaldehyde's contribution to urothelial dysfunction in vitro. *Arch Toxicol*, 2019. https://doi.org/10.1007/s00204-019-02589-1
11. M.A. Ayza, K.A. Zewdie, E.F. Yigzaw, S.G. Ayele, B.A. Tesfaye, G.G. Tafere, M.G. Abrha; Potential Protective Effects of Antioxidants against Cyclophosphamide-Induced Nephrotoxicity. *Int J Nephrol*. 2022 Apr. https://doi.org/10.1155/2022/5096825
12. Sherif IO. The effect of natural antioxidants in cyclophosphamide-induced hepatotoxicity: Role of Nrf2/HO-1 pathway. *Int Immunopharmacol*, 2018. https://doi.org/10.1016/j.intimp.2018.05.007
13. S.A.M. Saghir, S.A. Alharbi, M.A. Al-Garadi, N. Al-Gabri, H.Y. Rady, N.K. Olama, M.A.M. Abdulghani, A.M. Al Hroob, A.A. Almaiman, M. Bin-Jumah, et al.; Curcumin Prevents Cyclophosphamide-Induced Lung Injury in Rats by Suppressing Oxidative Stress and Apoptosis. *Processes*, 2020. https://doi.org/10.3390/pr8020127
14. E. Şengül, V. Gelen, S. Gedikli, S. Özkanlar, C. Gür, F. Çelebi, A. Çınar; The protective effect of quercetin on cyclophosphamide-Induced lung toxicity in rats. *Biomed Pharmacother*, 2017. https://doi.org/10.1016/j.biopha.2017.05.047

15. Y. A. Asiri; Probucol attenuates cyclophosphamide-induced oxidative apoptosis, p53 and Bax signal expression in rat cardiac tissues. *Oxidative Medicine and Cellular Longevity*, 2010. https://doi.org/10.4161/oxim.3.5.13107
16. S. K. Bjelogrlic, J. Radic, V. Jovic, S. Radulovic; Activity of d,l-alpha-Tocopherol (vitamin e) against cardiotoxicity induced by doxorubicin and doxorubicin with cyclophosphamide in mice; *Basic Clinical Pharmacology Toxicology*, 2005. https://doi.org/10.1111/j.1742-7843.2005.pto_166.x
17. A. Iqubal, S. Sharma, M. A. Ansari *et al*.; Nerolidol attenuates cyclophosphamide-induced cardiac inflammation, apoptosis and fibrosis in Swiss Albino mice. *European Journal of Pharmacology*, 2019. https://doi.org/10.1016/j.ejphar.2019.172666
18. S. H. Aladaileh, F. K. Al-Swailmi, M. H. Abukhalil, and M. H. Shalayel; Renoprotective effect of formononetin against cyclophosphamide-induced oxidative stress and inflammation in rat kidney. *Journal of Pharmaceutical Research International*, 2021. https://doi.org/10.9734/jpri/2021/v33i231144
19. P. Abraham and S. Rabi; Protective effect of aminoguanidine against cyclophosphamide-induced oxidative stress and renal damage in rats. *Redox Report*, 2011. https://doi.org/10.1179%2F174329211X12968219310837
20. D. Athira Mohan, V. Jain, and M. Jain; Evaluation of nephroprotective effect of rutin against toxic effect of cyclophosphamide. *Journal of Pharmacognosy and Phytochemistry*, 2020. https://doi.org/10.22271/phyto.2020.v9.i5y.12596
21. J. Wang, X. Tong, P. Li, H. Cao, W. Su; Immuno-enhancement effects of Shenqi Fuzheng Injection on cyclophosphamide-induced immunosuppression in Balb/c mice. *Journal of Ethnopharmacology*, 2012. https://doi.org/10.1016/j.jep.2011.12.019
22. Y. Mei, H. Chen, J. Zhang, X. Zhang, Y. Liang; Protective effect of chitooligosaccharides against cyclophosphamide-induced immunosuppression in mice. *International Journal of Biological Macromolecules*, 2013. https://doi.org/10.1016/j.-ijbiomac.2013.09.038
23. X. Chen, W. Nie, S. Fan, J. Zhang, Y. Wang, J. Lu, L. Jin; A polysaccharide from *Sargassum fusiforme* protects against immunosuppression in cyclophosphamide-treated mice. *Carbohydrate Polymers*, 2012. https://doi.org/10.-1016/j.carbpol.2012.06.052
24. N, Liu, Z. Dong, X. Zhu, H. Xu, Z. Zhao; Characterization and protective effect of *Polygonatum sibiricum* polysaccharide against cyclophosphamide-induced immunosuppression in Balb/c mice. *International Journal of Biological Macromolecules*, 2018. https://doi.org/10.1016/j.ijbiomac.2017.09.051
25. Q. Yu, S.-P. Nie, J.-Q. Wang, X.-Z. Liu, P.-F. Yin, D.-F. Huang, W.-J. Li, D.-M. Gong, M.-Y. Xie; Chemoprotective effects of *Ganoderma atrum* polysaccharide in cyclophosphamide-induced mice. *International Journal of Biological Macromolecules*, 2014. https://doi.org/10.1016/j.ijbiomac.2013.12.029
26. V.P. Kumar & Y. P. Venkatesh; Alleviation of cyclophosphamide-induced immunesuppression in Wistar rats by onion lectin (*Allium cepa* agglutinin). *Journal of Ethnopharmacology*, 2016. https://doi.org/10.1016/j.jep.2016.04.006
27. R.M. Medina-Sauza, M. Álvarez-Jiménez, A. Delhal, F. Reverchon, M. Blouin, J.A. Guerrero-Analco, C.R. Cerdán, R. Guevara, L. Villain, I. Barois; Earthworms Building

Up Soil Microbiota, a Review. *Front. Environ. Sci.*, 2019. https://doi.org-/10.3389/fenvs.2019.00081
28. M. Grdisa; Therapeutic properties of earthworms. *Biorem. Biodiv. Bioavail*, 2013.
29. D. Wang, Z. Ruan, R. Zhang, X. Wang, R. Wang, Z. Tang; Effect of Earthworm on Wound Healing: A Systematic Review and Meta-Analysis. *Front. Pharmacol*, 2021. https://doi.org/10.3389/fphar.2021.691742
30. P. Czaplewska, A. Bogucka, K. Macur, M. Rybicka, M. Rychłowski, M.J. Fiołka; Proteomic response of A549 lung cancer cell line to protein-polysaccharide complex Venetin-1 isolated from earthworm coelomic fluid. *Front Mol Biosci*, 2023. https://doi.org/10.3389/fmolb.2023.1128320
31. E. Kauschke, W. Mohrig, E. L. Cooper; Coelomic fluid proteins as basic components of innate immunity in earthworms. *European Journal of Soil Biology*, 2007. https://doi.org/10.1016/j.ejsobi.2007.08.043
32. H. Kobayashi, Y. Sekizawa, S. Shioda, S. Natori, T. Nakajima, M. Umeda; Lysenin, a Novel Bioactive Protein Isolated from Coelomic Fluid of the Earthworm Eisenia foetida - Structure, Secretion and Biological Activity. *Neuroendocrinology*, 1997. https://doi.org/10.1007/978-3-642-60915-2_19
33. Z. Hua, Y.H. Wang, H.W. Cao, L.J. Pu, Y.D. Cui; Purification of a protein from coelomic fluid of the earthworm *Eisenia foetida* and evaluation of its hemolytic, antibacterial, and antitumor activities. *Pharm Biol*, 2011. https://doi.org/10.3109-/13880209.2010.508498
34. P. Czaplewska, A. Bogucka, K. Macur , M. Rybicka, M.Rychłowski, M. J. Fiołka; Proteomic response of A549 lung cancer cell line to protein-polysaccharide complex Venetin-1 isolated from earthworm coelomic fluid. *Fron. Mol Biosci*, 2023. https://doi.org/10.3389/fmolb.2023.1128320
35. M. Rybicka, P. Czaplewska, J. Rzymowska, W. Sofińska-Chmiel, S. Wójcik-Mieszawska, K. Lewtak, K. Węgrzyn, P. Jurczak, A. Szpiech, J. Nowak, N. Musiał, M. J. Fiołka; Novel Venetin-1 nanoparticle from earthworm coelomic fluid as a promising agent for the treatment of non-small cell lung cancer. *Sci Rep*, 2022. https://doi.org/10.1038/s41598-022-21665-8
36. S. Wójcik-Mieszawska, K. Lewtak, E. Skwarek, D. Dębowski, A. Gitlin-Domagalska, J. Nowak, J. Wydrych, J. Pawelec, M. J. Fiołka; Autophagy of *Candida albicans* cells after the action of earthworm Venetin-1 nanoparticle with protease inhibitor activity. *Sci Rep*, 2023. https://doi.org/10.1038/s41598-023-41281-4
37. S. Wójcik-Mieszawska, K. Lewtak, W. Sofińska-Chmiel, J. Wydrych, M. J. Fiołka; Atypical changes in *Candida albicans* cells treated with the Venetin-1 complex from earthworm coelomic fluid. *Sci Rep*, 2023. https://doi.org/10.1038/s41598-023-29728-0
38. M. Mizerska-Kowalska, B. Zdzisińska, M. Fiołka, K. Musiał; Protein-sugar fraction isolated from the coelomic fluid of Dendrobaena veneta earthworm for application in non-specific immunotherapy. Patent PL 238444, P.429033; 25.02.2019.
39. J. R. Wiśniewski, A. Zougman, N. Nagaraj, M. Mann; Universal sample preparation method for proteome analysis. *Nat Methods*, 2009. https://doi.org/10.1038/nmeth.1322

40. J. Rappsilber, M. Mann, Y. Ishihama; Protocol for micro-purification, enrichment, pre-fractionation and storage of peptides for proteomics using StageTips. *Nat Protoc*, 2007. https://doi.org/10.1038/nprot.2007.261
41. S.H. Yu, D. Ferretti, J.P. Schessner, J.D. Rudolph, G.H.H. Borner, J.Cox; Expanding the Perseus Software for Omics Data Analysis With Custom Plugins. *Curr Protoc Bioinformatics*, 2020. https://doi.org/10.1002/cpbi.105
42. D. Szklarczyk, J.H. Morris, H. Cook, M. Kuhn, S. Wyder, M. Simonovic, A. Santos, N.T. Doncheva, A. Roth, P. Bork, L.J. Jensen, C. von Mering; The STRING database in 2017: quality-controlled protein–protein association networks, made broadly accessible. *Nucleic Acids Research*, 2017. https://doi.org/10.1093/nar/gkw937
43. M. Kanehisa, S. Goto, Y. Sato, M. Furumichi, M. Tanabe; KEGG for integration and interpretation of large-scale molecular data sets. *Nucleic Acids Res*, 2012. https://doi.org/10.1093/nar/gkr988
44. B. Li, W. Shen, H. Peng, Y. Li, F. Chen, L. Zheng, J. Xu, L. Jia; Fibronectin 1 promotes melanoma proliferation and metastasis by inhibiting apoptosis and regulating EMT. *Onco Targets Ther*, 2019. https://doi.org/10.2147/OTT.S195703
45. A. Hoter, M.E. El-Sabban, H.Y. Naim; The HSP90 Family: Structure, Regulation, Function, and Implications in Health and Disease. *Int J Mol Sci*, 2018. https://doi.org/10.3390/ijms19092560
46. N.H. Sulimai, J. Brown, D. Lominadze; Fibrinogen, Fibrinogen-like 1 and Fibrinogen-like 2 Proteins, and Their Effects. *Biomedicines*, 2022. https://doi.org/10.3390-/biomedicines10071712
47. H. Kimura, K. Fumoto, K. Shojima, S. Nojima, Y. Osugi, H. Tomihara, H. Eguchi, Y. Shintani, H. Endo, M. Inoue, Y. Doki, M. Okumura, E. Morii, A. Kikuchi; CKAP4 is a Dickkopf1 receptor and is involved in tumor progression. *J Clin Invest*, 2016. https://doi.org/10.1172/JCI84658
48. P. Xia, X. Ma, L. Yan, S. Lian, X. Li, Y. Luo, Z. Chen, X. Ji; Generation and Application of Monoclonal Antibodies against Porcine S100A8, S100A9, and S100A12 Proteins Using Hybridoma Technology. *Int J Mol Sci*, 2024. https://doi.org/10.1111/imm.13722
49. Z. Li, J. Liu, Z. Liu, X. Zhu, R. Geng, R. Ding, H. Xu, S. Huang; Comprehensive analysis identifies crucial genes associated with immune cells mediating progression of carotid atherosclerotic plaque. *Aging*, 2024. https://doi.org/10.18632/aging.205566
50. A. Zehrer, R. Pick, M. Salvermoser, A. Boda, M. Miller, K. Stark, L.T. Weckbach, B. Walzog, D. Begandt; A Fundamental Role of Myh9 for Neutrophil Migration in Innate Immunity. *J Immunol*, 2018. https://doi.org/10.4049/jimmunol.1701400
51. J.F. Neves, J. Zhu, P. Sousa-Victor, M.J. Konjikusic, R.R. Riley, S. Chew, Y. Qi, H. Jasper, D.A. Lamba; Immune modulation by MANF promotes tissue repair and regenerative success in the retina. *Science*, 2016. https://doi.org/1010.1126/-science.aaf3646
52. A. Yusufujiang, S. Zeng, C. Yang, S. Jing, L. Yang, H. Li; PCBP-1 Regulates the Transcription and Alternative Splicing of Inflammation and Ubiquitination-Related Genes in PC12 Cell. *Front Aging Neurosci*, 2022. https://doi.org/10.3389/fnagi.-2022.884837

53. J-S. Kim, Y-C. Lee, H-T. Nam, G. Li, E-J. Yun, K-S. Song, K-S. Seo, J-H. Park, J-W. Ahn, O. Zee, J-I. Park, W-H. Yoon, K. Lim, B-D. Hwang; Apicularen A Induces Cell Death through Fas Ligand Up-Regulation and Microtubule Disruption by Tubulin Down-Regulation in HM7 Human Colon Cancer Cells. *Clin Cancer Res*, 2007. https://doi.org/10.1158/1078-0432.CCR-07-1428
54. N. Benavides and C.G. Giraudo; Extended-Synaptotagmin-1 and -2 control T cell signaling and function. EMBO Rep, 2023. https://doi.org/10.1038/s44319-023-00011-7
55. P. Guo, Y. Liu, F. Geng, A.W. Daman, X. Liu, L. Zhong, A. Ravishankar, R. Lis, J.G. Barcia Durán, T. Itkin, F. Tang, T. Zhang, J. Xiang, K. Shido, B.S. Ding, D. Wen, S.Z. Josefowicz, S. Rafii; Histone variant H3.3 maintains adult haematopoietic stem cell homeostasis by enforcing chromatin adaptability. *Nat Cell Biol*, 2022. https://doi.org/10.1038/s41556-021-00795-7
56. K.A. Wierenga, F.M.. Riemers, B Westendorp, J.R. Harkema, J.J. Pestka; Single cell analysis of docosahexaenoic acid suppression of sequential LPS-induced proinflammatory and interferon-regulated gene expression in the macrophage. *Front Immunol*, 2022. https://doi.org/10.3389/fimmu.2022.993614
57. L. Shijie, P. Zhen, Q. Kang, G. Hua, Y. Qingcheng, C. Dongdong; Deregulation of CLTC interacts with TFG, facilitating osteosarcoma via the TGF-beta and AKT/mTOR signaling pathways. *Clinical and Translational Medicine*, 2011. https://doi-.org/10.1002/ctm2.377
58. X. Liu, Z. Lin, T. Zhou, R. Zong, H. He, Z. Liu, J.X. Ma, Z. Liu, Y. Zhou. Anti-angiogenic and anti-inflammatory effects of SERPINA3K on corneal injury. *PLoS One*, 2011. https://doi.org/10.1371/journal.pone.0016712
59. J. Li, P. Fang, Y. Zhou, D. Wang, L. Fang, S. Xiao; DEAD-box RNA helicase 21 negatively regulates cytosolic RNA-mediated innate immune signaling. *Front. Immunol*, 2022. https://doi.org/10.3389/fimmu.2022.956794
60. J. Zan, R. Xu, X. Tang, M. Lu, S. Xie, J. Cai, Z. Huang, J. Zhang; RNA helicase DDX5 suppresses IFN-I antiviral innate immune response by interacting with PP2A-Cβ to deactivate IRF3. *Exp Cell Res*, 2020. https://doi.org/10.1016/j.yexcr.2020.112332
61. U. Anand, A. Dey, A.K.S. Chandel, R. Sanyal, A. Mishra, D.K. Pandey, V. De Falco, A. Upadhyay, R. Kandimalla, A. Chaudhary, J.K. Dhanjal, S. Dewanjee, J. Vallamkondu, J.M. Pérez de la Lastra; Cancer chemotherapy and beyond: Current status, drug candidates, associated risks and progress in targeted therapeutics. *Genes Dis*, 2022. https://doi.org/10.1016/j.gendis.2022.02.007
62. M. Sileghem, J.N. Flynn, A. Darji, P. De Baetselier, J. Naessens; Parasitic Infections and the Immune System | African Trypanosomiasis. *Parasitic Infections and the Immune System*, 1994. https://doi.org/10.1016/B978-0-08-092405-2.50005-9
63. A.H. Bhat, K.B. Dar, S. Anees, M.A. Zargar, A. Masood, M.A. Sofi, S.A. Ganie; Oxidative stress, mitochondrial dysfunction and neurodegenerative diseases; a mechanistic insight. *Biomed Pharmacother*, 2015. https://doi.org/10.1016/j.biopha.-2015.07.025